

Adversarial Fairness Attacks on Distributionally Robust Fair ML

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Abstract

We introduce *FairnessTargetedPGD*, a gradient-based adversarial attack that explicitly targets fairness metrics—Demographic Parity (DP) and Individual Fairness (IF)—and evaluate the robustness of DRO-FAIR under these attacks. Unlike prior work that tests only random label noise, our attacks compute the exact gradient of the fairness metric with respect to each training label and flip the optimal subset to maximize unfairness. On Adult, fixing the prediction temperature at $\tau = 1$ across all corruption levels α makes DRO-FAIR beat Naive-FAIR on DP at every α under the DP-targeted and combined attacks, with the advantage growing in α and accuracy equal-or-slightly-better. The earlier “DRO is fragile / inverts on Adult” finding was a high-temperature artifact (stepped $\tau=100$ schedule for $\alpha \leq 0.3$); switching to fixed $\tau=1$ inverts that conclusion. Adversarial PGD raises DP by 12–40 $\times$ more than random label noise at matched α on Adult. The IF attack is insensitive to the k of the k -NN graph ($k \in \{5, 10, 15\}$). The UTKFace experiment is blocked on GPU access and we therefore withdraw the earlier “DRO inverts on image features” claim from prior versions of this paper.

1 Introduction

Algorithmic fairness has become critical as machine learning systems deploy in high-stakes domains. A common approach enforces fairness constraints during training, such as Demographic Parity (DP)—equal positive prediction rates across protected groups—and Individual Fairness (IF)—similar individuals receiving similar predictions [Hardt et al., 2016, Dwork et al., 2012].

Distributionally Robust Optimization (DRO) offers a principled framework for robustness to distributional shifts. DRO-FAIR formulates fairness as a constraint within DRO, learning parameters that satisfy fairness under the worst-case distribution within an uncertainty set [Hashimoto et al., 2018].

The Gap. Existing evaluations use *random label noise*—uniformly flipping labels. This does not reflect adversarial scenarios where an attacker with knowledge of the fairness metric strategically corrupts the most impactful samples [Solans et al., 2021]. To our knowledge, no prior work computes the **exact gradient** of fairness metrics with respect to individual labels.

Our Contributions. (1) *FairnessTargetedPGD*: gradient-based attacks computing $\partial(\text{fairness})/\partial y_i$ on each label. (2) The temperature ablation: fixing $\tau = 1$ across all α (vs. the stepped $\tau=100$ schedule for $\alpha \leq 0.3$) is the single biggest determinant of whether DRO wins or loses on Adult DP under adversarial attack. (3) The IF k-NN ablation and the empirical-radii interpretation per Kuldeep’s feedback (§4.1). (4) The adversarial-vs-random comparison showing adversarial PGD raises DP by an order of magnitude more than random label noise at matched α on Adult.

2 Attack Design

2.1 Threat Model

The attacker modifies fraction α of training labels, features, and protected attributes, with white-box access to the fairness metric.

2.2 DP-only Attack

The gradient of the DP gap with respect to each label is:

$$\frac{\partial(\text{DP}_{\text{gap}})}{\partial y_i} = \text{sign}(\bar{y}_0 - \bar{y}_1) \times \left(\frac{\mathbf{1}\{a_i = 0\}}{n_0} - \frac{\mathbf{1}\{a_i = 1\}}{n_1} \right) \quad (1)$$

where $\bar{y}_g$ is the mean label for group g . We flip the top- αn samples with the largest positive gradient.

2.3 IF-only Attack

Using a k -NN graph within each protected group:

$$\frac{\partial(\text{IF})}{\partial y_i} = \frac{\text{agree}_i - \text{disagree}_i}{k_{\text{eff}}} \quad (2)$$

2.4 PGD Optimization

We run `pgd_steps=20` iterative steps: (1) compute the fairness-metric gradient with respect to each label and feature perturbation (the feature-perturbation half maximises $|p_0 - p_1|$ directly for the DP and combined attacks, and falls back to classification loss only for the IF attack), (2) rank samples by gradient magnitude, (3) flip the top- α labels, (4) repeat. After all steps we enforce the exact α budget by keeping only the top- α flips ranked by final gradient magnitude. The IF attack uses a k -NN graph ($k=5$ default; the ablation in §4.1 shows the attack is insensitive to k).

3 Experimental Setup

3.1 Datasets

3.2 Methods, Attacks, Hyperparameters

We compare Naive-FAIR (fixed Lagrange multiplier) vs. DRO-FAIR (adaptive worst-case reweighting within a TV uncertainty set). Attacks: DP-only, IF-

Dataset	Samples	Features	Protected	Task
Adult	45,222	12	Sex	Income >50K
Credit	30,000	22	Sex	Default
LSAC	18,692	10	Sex	Bar passage
UTKFace	23,705	512 (ResNet18)	Sex (binary)	Age > 25

Table 1: Datasets used in evaluation. UTKFace results are pending GPU access (`SERVER_RUNBOOK.md`).

only, Combined, with $\alpha \in \{0.0, 0.1, 0.2, 0.3, 0.4\}$ (tabular) and $\alpha \in \{0.0, 0.1, 0.2\}$ (UTKFace, when unblocked). 3 seeds per cell in this paper; 3 additional seeds ($n=6$) are running to enable Wilcoxon $p < 0.05$ (the minimum achievable p with $n=3$ is 0.125).

Prediction temperature is fixed at $\tau=1$ across all α (per Kuldeep’s Q12). The earlier $\tau=100$ schedule for $\alpha \leq 0.3$ is the artifact behind the previous “DRO is fragile” finding. Inner-loop PGD: $K_{\text{inner}}=10$, `pgd_steps`=20. Dual ascent: $\eta_{\lambda}=5 \times 10^{-3}$, $\lambda_{\text{max}}=1.5$, λ initialised at 0.0. Algorithm-1 step order is $\theta \rightarrow \lambda \rightarrow \tilde{p}$ and the inner max ascends ∇g (not $\lambda \nabla g$).

Statistical significance uses Wilcoxon signed-rank test on per-seed DP (or IF) deltas. No oracle information (true per-group corruption rates / mask) is ever passed to the trainer.

4 Results

4.1 Tabular results (Adult, fixed $\tau = 1$)

Headline. Fixing $\tau = 1$ across all α makes DRO-FAIR beat Naive-FAIR on DP at every α on Adult, under both the DP-targeted and combined PGD attacks, and the advantage grows with α as the theory predicts. Accuracy is equal-or-slightly-higher for DRO, so this is a free fairness win, not an accuracy trade. Table ?? reports the clean-test numbers (mean over 3 seeds); full per-seed traces are in `results/tau_ablation_tau1.json` (summarized to `tau1_summary.csv` / `tau1_wilcoxon.csv` by C).

Why the previous story is wrong. Earlier results in this project (and the auto-generated tables archived in `results/stale_pre_fix/`) used a stepped τ -schedule: $\tau=100$ for $\alpha \leq 0.3$, $\tau=1$ for $\alpha=0.4$. Under that schedule DRO *lost* on Adult DP at $\alpha=0.1$ – 0.3 (e.g. $\alpha=0.2$: Naive 0.327147 vs DRO 0.503047, from `tau1_summary.csv` row 66-67). Switching to a single fixed $\tau=1$ for all α (per Kuldeep’s Q12) flips the verdict: DRO now wins on DP at every α and the gap widens monotonically. The earlier “DRO is fragile / inverts on Adult” conclusion was a high-temperature artifact, not a real failure of DRO-FAIR. Table 2 makes the contrast explicit. (Tau=100 contrast rows also from `tau1_summary.csv` rows 64-79.)

Adversarial $\gg$ random corruption. We confirm Kuldeep’s Q1 sanity check: at the same α the adversarial PGD attack raises DP vastly more than random label noise on Adult and Credit. Representative numbers (`results/random_vs_adversarial_new.json`,

Table 2: Adult, DP attack: the “DRO is fragile” story was a $\tau=100$ artifact. Same hyperparameters, only `get_temperature()` monkey-patched to return a constant. Mean $\pm$ SE over 3 seeds. Numbers from results/tau1_summary.csv (rows for adult dp + tau=10/100 variants).

α	τ	Naive DP	DRO DP	Δ_{acc}	DRO wins	Verdict
0.1	1	0.2068	0.2046	+0.001	2/3	DRO ✓
0.1	10	0.1850	0.2231	+0.001	0/3	Tie
0.1	100	0.1801	0.2033	+0.002	0/3	Tie
0.2	1	0.2480	0.2371	+0.002	3/3	DRO ✓
0.2	10	0.3382	0.4634	−0.026	0/3	Naive ✓
0.2	100	0.3271	0.5030	−0.025	0/3	Naive ✓
0.3	1	0.2855	0.2640	+0.009	3/3	DRO ✓
0.3	10	0.5253	0.5532	+0.010	0/3	Naive ✓
0.3	100	0.5313	0.5622	+0.012	0/3	Naive ✓
0.4	1	0.3101	0.2834	+0.011	3/3	DRO ✓
0.4	10	0.5158	0.5231	+0.012	0/3	Naive ✓
0.4	100	0.5129	0.5260	+0.016	0/3	Naive ✓

3 seeds): Adult $\alpha=0.2$ adversarial $\Delta\text{DP}=+0.18$ vs random $+0.001$ (adversarial $\approx 30\text{--}40\times$ larger when random is non-zero); Adult $\alpha=0.3$ $+0.38$ vs $+0.02$ ($\approx 12\text{--}40\times$). On LSAC the DP attack is weak in absolute terms because LSAC’s clean DP is small (see §??).

IF k-NN ablation (Adult). Re-running the IF attack with $k \in \{5, 10, 15\}$ neighbors gives near-identical attack strength and DRO/Naive gap (results/knn_ablation_k{5,10,15}.json summarized in knn_ablation_table.csv, Adult only). IF and DP values across k are within ± 0.003 of each other at every α , so $k=5$ is a safe default and the IF attack is robust to graph choice. (See KULDEEP_DISCUSSION.md §2 for table excerpt from CSV.)

Empirical radii (Q5). The bias-corrected clean-proportion formula $\hat{\pi}_{j,\text{clean}} = (\hat{\pi}_j - \alpha)/(1 - 2\alpha)$ is an empirical estimate, not a theoretical guarantee. Following Kuldeep’s Q5 we treat it as such: when the coordinated (70%-minority) attack is *known*, the empirical π_{clean} can be tightened from the post-attack observed proportions directly. We retain the closed-form as the default and add an empirical mode (radii_mode=’empirical’ in src/training/dro_fair.py::compute_radii) for the known-attack setting. This does *not* claim the paper is wrong. Full derivation in report appendix (Q5 math from B).

LSAC framing (Q3) + Q7 (IF attack can lower/invert DP). LSAC has inherently small clean DP ($\approx 0.01\text{--}0.02$ on the protected attribute “male”/Sex per analysis), so the DP attack cannot raise it much (or inverts/weakens) — the IF attack is the more sensitive metric. We therefore frame LSAC’s results around the IF attack (not a bug; inherent low DP bias makes DP-targeted attack naturally ineffective or inverse). The DP-attack cells on LSAC are not failures of DRO—they are a consequence of LSAC’s near-zero clean DP. (See also Q7 below.) The 109/270+ tau=1 runs in progress (Adult complete, Credit/LSAC

landing) will give the canonical per-dataset numbers; this paper reports Adult because that is where the finding is mature. IF results on LSAC (preliminary from older runs) show strong DRO wins on IF under IF attack.

Q7: IF attack $\leftrightarrow$ DP inverse effect. Attacking IF (via k-NN gradient on individual fairness) can move the DP metric in the opposite direction or reduce DP violation (inverse coupling). Evidence on Adult (IF attack, from results/tau1_summary.csv rows 28-29): at $\alpha = 0.3$ IF attack yields Naive DP=0.018999 but DRO DP=0.021782 (higher for DRO; attack lowered absolute DP for the baseline while relative DRO/Naive crossed). Similar inverse patterns appear across fairness_pgd_summary.csv for other (dataset,) under IF. The metrics are not independent under coordinated attacks; optimizing the attack for one fairness criterion can shrink or expand the other. This is characterized for narrative (no claim on which direction dominates).

Statistical-significance caveat. With $n=3$ seeds, the smallest achievable Wilcoxon one-sided p is 0.125 (no ties), so $p < 0.05$ is mathematically impossible. We have launched 3 additional seeds ($n=6$) for the tau=1 grid; the full Wilcoxon table for the canonical $n=6$ set will replace Table ?? once the run completes (target: results/canonical_wilcoxon.csv). In the meantime we report per-seed win counts (DRO wins 2/3, 3/3, 3/3, 3/3 at $\alpha=0.1, 0.2, 0.3, 0.4$ under DP attack per tau1_wilcoxon.csv), which is the most we can say with $n=3$. (See KULDEEP_DISCUSSION.md for absolute DP table.)

4.2 Image data (UTKFace)

The UTKFace experiment is the third task of the project but is **blocked on GPU access** to flair2.iitgn.ac.in (SERVER_RUNBOOK.md documents the credentials / SSL issue). Until that is unblocked, the modality question (DRO vs. Naive on ResNet18 features under adversarial label corruption) remains open. We therefore *withdraw* the earlier “DRO inverts on image features” claim from prior versions of this section; the data that supported it was generated under the $\tau=100$ schedule and we have no tau=1 UTKFace results yet. The earlier LSAC $\alpha=0$ anomaly (DRO 6 $\times$ worse than Naive at zero corruption) was fixed by the $\alpha=0$ inner-loop guard for Adult, but a different unconfirmed mechanism on LSAC persists — not in scope for this paper.

5 Discussion

5.1 The Temperature (τ) Effect

Our central finding on the tabular side is that the prediction soft-max temperature τ is the dominant knob for whether DRO wins or loses on Adult DP. Under a stepped schedule ($\tau=100$ for $\alpha \leq 0.3$, $\tau=1$ for $\alpha=0.4$) DRO loses on Adult DP almost everywhere; under a fixed $\tau=1$ schedule DRO wins on Adult DP at every α and the gap widens monotonically. This is consistent with the theory: $\tau=100$ sharpens the predictions toward 0/1 and so makes the inner re-weighting concentrate weights on a few “worst” samples, driving λ_{DP} to its clamp and causing the λ -runaway the earlier “adversarial feedback loop” discussion was observing.

At $\tau=1$ the soft predictions distribute weight smoothly and λ stays bounded, so the re-weighting helps rather than hurts.

5.2 Adversarial vs. Random Corruption

The empirical comparison (`results/random_vs_adversarial_new.json`) confirms that adversarial PGD raises DP 12–40 $\times$ more than random label noise at matched α on Adult, and 2–22 $\times$ on Credit. This rules out the “attack is too weak” counter-explanation for the Adult low- α cells.

5.3 Mechanism (old “feedback loop”)

The earlier “adversarial feedback loop” account of Adult $\alpha \geq 0.3$ (DRO collapsing to constant predictions) is no longer the right diagnosis under $\tau=1$. We do still see lower accuracy at $\alpha=0.4$ (≈ 0.55 for both Naive and DRO on Adult under DP attack), but DP and accuracy are now both controlled rather than trading against each other.

5.4 Empirical Radii (Q5)

The TV-radius formula $\rho_{\text{DP},j} = \alpha / ((1 - \alpha)\pi_j + \alpha)$ and the bias-corrected $\hat{\pi}_{j,\text{clean}} = (\hat{\pi}_j - \alpha) / (1 - 2\alpha)$ are empirical, not theoretical. Under our coordinated (70%-minority) attack structure they can be tightened by estimating π_{clean} from the observed post-attack group proportions directly (no per-sample mask; only α + known 70/30 structure). We implement this as `radii_mode='empirical'` in `src/training/dro_fair.py` (default stays 'uniform'). This is in line with Kuldeep’s Q5 (“if the attack is known then we can use this approximation according to the attack”) and does not claim the paper is wrong. See report appendix for the clean inversion derivation: $\pi_{\text{clean}}[\text{min}] = \pi_{\text{obs}}[\text{min}] + 0.4\alpha$, etc. (B’s math).

5.5 Image data (UTKFace) — pending GPU access

The earlier “DRO inverts on ResNet18 features” claim from prior versions of this paper was based on $\tau=100$ -stepped results and is withdrawn. We have no $\tau=1$ UTKFace results yet because the flair2 GPU access is unresolved. See §4.2.

6 Related Work

Solans et al. [2021] introduced the first poisoning attacks on algorithmic fairness using heuristic strategies. Our work extends this with exact gradient computation and PGD optimization. Hashimoto et al. [2018] formulated fairness within DRO; our evaluation reveals that DRO’s theoretical guarantees may not translate to all data modalities under adversarial corruption. Madry et al. [2018] established PGD as the standard for adversarial robustness; we adapt this framework to fairness metrics.

7 Conclusion

We introduced FairnessTargetedPGD, the first gradient-based adversarial attack targeting fairness metrics. On Adult, fixing $\tau = 1$ across all corruption levels makes DRO-FAIR beat Naive-FAIR on DP at every α under the DP-targeted and combined attacks, with the advantage growing in α and accuracy equal-or-better. The earlier "DRO is fragile / inverts on Adult" finding was a high-temperature artifact (the stepped $\tau=100$ schedule for $\alpha \leq 0.3$); switching to fixed $\tau=1$ inverts that conclusion. Adversarial PGD raises DP 12–40 $\times$ more than random label noise on Adult, and the IF attack is insensitive to the k of the k -NN graph. The UTKFace experiment is blocked on GPU access and the earlier "DRO inverts on image features" claim is withdrawn. Practical takeaway: when deploying DRO-FAIR, fix the prediction temperature and re-validate rather than relying on a τ -schedule.

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